

point-identification. We say that g is scale-invariant if its value is the same under any re-scaling of the units of y by a positive constant a .

Definition 1. We say that the function g is *scale-invariant* if it is homogeneous of degree zero, i.e. $g(y_1, y_0) = g(ay_1, ay_0)$ for all $a, y_1, y_0 > 0$.

We next describe point-identification. We consider parameters that are identified without placing restrictions on treatment effect heterogeneity. As in [Fan et al. \(2017\)](#), this is formalized by considering parameters that can be learned if we know the marginal distributions of $Y(1)$ and $Y(0)$, but not the full joint distribution of $(Y(1), Y(0))$.

To connect treatment effect heterogeneity to the joint distribution of potential outcomes, consider the simple case of a randomized experiment. By examining the outcome distribution for the treated group, we can learn the marginal distribution of $Y(1)$. Likewise, by examining the outcome distribution for the control group, we can learn the marginal distribution of $Y(0)$. If treatment effects were assumed to be constant, then for each observed treated unit with outcome $Y(1)$, we could infer their untreated outcome as $Y(0) = Y(1) - \tau$, where τ is the average treatment effect. Hence, the joint distribution of $(Y(1), Y(0))$ would be identified. However, if we allow for treatment effect heterogeneity, then for an observed treated unit with outcome $Y(1)$, we do not know what their value of $Y(0)$ would be, and thus we do not know the joint distribution of $(Y(1), Y(0))$. This winds up being especially important in settings with an extensive margin, since when we observe the distribution of outcomes for treated units, it means that we do not know *which* of the treated units would have had a zero outcome under the control condition, and thus it is difficult to disentangle the intensive and extensive margins.¹⁷

With that intuition in mind, we now give a formal definition. Recall that P denotes the joint distribution of $(Y(1), Y(0))$, while $P_{Y(d)}$ denotes the marginal distribution of $Y(d)$. We then say θ_g is point-identified if it depends on P only through the marginals $P_{Y(1)}, P_{Y(0)}$.

Definition 2 (Identification). We say that θ_g is *point-identified from the marginals at P* if for every joint distribution Q with the same marginals as P (i.e. such that $Q_{Y(d)} = P_{Y(d)}$ for $d = 0, 1$), $E_P[g(Y(1), Y(0))] = E_Q[g(Y(1), Y(0))]$. For a class of distributions $\mathcal{P}$, we say that θ_g is *point-identified over $\mathcal{P}$* if for every $P \in \mathcal{P}$, θ_g is point-identified from the marginals at P .

We will denote by $\mathcal{P}_+$ the set of distributions on $[0, \infty)^2$. Thus, θ_g is point-identified over $\mathcal{P}_+$ if it is always identified when Y takes on zero or weakly positive values. Our next result formalizes that it is not possible to have a parameter of the form $E_P[g(Y(1), Y(0))]$ that is both scale-invariant and point-identified over $\mathcal{P}_+$.

Proposition 2 (A trilemma). *The following three properties cannot hold simultaneously:*

- (a) $\theta_g = E_P[g(Y(1), Y(0))]$ for a non-constant function $g : [0, \infty)^2 \rightarrow \mathbb{R}$ that is weakly increasing in its first argument.

¹⁷In [Appendix C](#), we discuss a variety of structural approaches that impose assumptions restricting the joint distribution, thus allowing us to separately point-identify the effects for the two margins.

(b) The function g is scale-invariant.

(c) θ_g is point-identified over $\mathcal{P}_+$.¹⁸

Any parameter defined with zero-valued outcomes must therefore abandon one of properties (a)–(c).

As a special case, [Proposition 2](#) implies that the ATE for any increasing function $m(Y)$ defined at zero cannot be scale-invariant. This is because the ATE for $m(Y)$ takes the form in (a) with $g(y_1, y_0) = m(y_1) - m(y_0)$, and is also point-identified (part (c)). It follows that property (b) must be violated, i.e. there is some $c, y_0, y_1 > 0$ such that $m(cy_1) - m(cy_0) \neq m(y_1) - m(y_0)$. [Proposition 2](#) thus formalizes the sense in which it is not possible to “fix” the issues with ATEs for log-like transformations described above by taking alternative transformations of the outcome (e.g. $\sqrt{Y}$).

3.1 Implications for settings without an extensive margin

The trilemma in [Proposition 2](#) applies for transformations of the outcome defined at zero. To prove [Proposition 2](#), however, we establish an even stronger result: the only parameter satisfying properties (a) and (b) that is point-identified over distributions for which Y is *strictly* positively-valued is the ATE in logs.¹⁹ This result, which is formalized in [Proposition 3](#) in the Appendix, has some useful implications for settings in which the outcome is strictly positive.

First, it implies that the ATE for any transformation of the outcome other than $\log(Y)$ will depend on the units of the outcome for at least some DGP where the outcome is strictly positive. The scale-dependence of log-like transformations such as $\log(1 + Y)$ or $\operatorname{arcsinh}(Y)$ is thus not entirely limited to settings with an extensive margin.²⁰ We note, however, that while the ATE for such transformations may depend on the units of the outcome even without zero-valued outcomes, the dependence need not be arbitrarily bad in the sense of [Proposition 1](#). Indeed, (2) shows that if there is no extensive margin, the ATE for a log-like transformation will be approximately insensitive to scaling once the values of Y are made large. This is intuitive, since if Y is strictly positively-valued, the ATE for a log-like transformation will be approximately equal to the ATE in logs when the values of Y are made large.

Second, [Proposition 3](#) implies that even when $Y(1)$ and $Y(0)$ are strictly-positively valued, the average proportional effect $\theta_{\text{Avg}\%} = E[(Y(1) - Y(0))/Y(0)]$ is not point-identified. This parameter is empirically relevant: For instance, [Andrews and Miller \(2013\)](#) show that in the [Bailey \(1978\)–Chetty \(2006\)](#) model with heterogeneous consumption responses to unemployment, the optimal level of unemployment insurance depends on a parameter of the form $\theta_{\text{Avg}\%}$, where Y is consumption and

¹⁸A minor technical complication arises from the fact that $E_P[g(Y(1), Y(0))]$ could be infinite for some P . For the purposes of our result, it suffices to trivially define θ_g to be identified in this case. Alternatively, the same result holds if part (c) is modified to impose only that θ_g is point-identified over all distributions in $\mathcal{P}_+$ with finite support, thus avoiding issues related to undefined expectations.

¹⁹More precisely, the only such treatment effect is the ATE in logs or an affine transformations thereof.

²⁰There is thus no conflict between our results and those in [Thakral and Tô \(2023\)](#), who note that semi-elasticities for OLS regressions using log-like transformations may depend on the units of the outcome even when Y is strictly positively-valued.

D is unemployment. Although the ATE in logs may approximate $\theta_{\text{Avg\%}}$ when the proportional effect of the treatment is approximately constant, our results imply that it is not possible to point-identify $\theta_{\text{Avg\%}}$ when allowing for arbitrarily heterogeneous proportional effects.

4 Empirical approaches with zero-valued outcomes

Our theoretical results above imply that when there are zero-valued outcomes, the researcher should not take a log-like transformation of the outcome and interpret the resulting ATE as an average percentage effect: Unlike a percentage, such an ATE depends on the units of the outcome. In this section, we highlight some other parameters that are well-defined and easily interpreted when there are zero-valued outcomes; in [Section 5](#) below, we show how these parameters can be estimated in three empirical applications. Of course, any alternative parameter must necessarily drop one of the requirements in the trilemma in [Proposition 2](#), but the choice of which to drop may depend on the researcher’s motivation.

To inform our discussion of alternative parameters, it is therefore useful to first enumerate several reasons why empirical researchers may target treatment effects for a log-transformed outcome rather than the ATE in levels:

- (i) The researcher is interested in reporting a treatment effect parameter with easily-interpretable units, such as “percentage changes.”
- (ii) The researcher believes that there are decreasing returns to the outcome, and thus wants to place more weight on treatment effects for individuals with low initial outcomes. For instance, the researcher may perceive it to be more meaningful to raise income from $Y(0) = \$10,000$ to $Y(1) = \$20,000$ than from $Y(0) = \$100,000$ to $Y(1) = \$110,000$, yet both of these treatment effects contribute equally to the ATE in levels.
- (iii) The researcher is interested in both the intensive and extensive margin effects of the treatment, and is using the ATE for a log-like transformation as an approximation to the proportional effect along the intensive margin.

These three motivations suggest different ways of breaking out of the trilemma in [Proposition 2](#). If the goal is to achieve a percentage interpretation, then one can consider scale-invariant parameters outside of the class $E_P[g(Y(1), Y(0))]$. For instance, researchers can consider the ATE in levels expressed as a percentage of the control mean, or the ATE for a normalized parameter $\tilde{Y}$ that already has a percentage interpretation. Alternatively, if the goal is to capture concave social preferences over the outcome, then it is natural to specify how much we value the intensive margin relative to the extensive margin—thus abandoning scale-invariance. Finally, if the goal is to separately understand the intensive margin effect, the researcher can abandon point-identification (from the marginal distributions) and directly target the partially identified parameter $E[\log(Y(1)) - \log(Y(0)) \mid Y(0) > 0, Y(1) > 0]$, the effect in logs for individuals with positive

outcomes under both treatments. We address each of these cases in turn below, with a summary of some possible parameters in Table 2.

Description	Parameter	Main property sacrificed?	Pros/Cons
Normalized ATE	$E[Y(1) - Y(0)]/E[Y(0)]$	$E[g(Y(1), Y(0))]$	<i>Pro</i> : Percent interpretation <i>Con</i> : Does not capture decreasing returns
Normalized outcome	$E[Y(1)/X - Y(0)/X]$	$E[g(Y(1), Y(0))]$	<i>Pro</i> : Per-unit- X interpretation <i>Con</i> : Need to find sensible X
Explicit tradeoff of intensive/extensive margins	ATE for $m(y) = \begin{cases} \log(y) & y > 0 \\ -x & y = 0 \end{cases}$	Scale-invariance	<i>Pro</i> : Explicit tradeoff of two margins <i>Con</i> : Need to choose x ; Monotone only if support excludes $(0, e^{-x})$
Intensive margin effect	$E\left[\log\left(\frac{Y(1)}{Y(0)}\right) \mid Y(1) > 0, Y(0) > 0\right]$	Point-identification	<i>Pro</i> : ATE in logs for the intensive margin <i>Con</i> : Partial identification

Table 2: Summary of alternative target parameters

Remark 8 (Statistical reasons for transforming the outcome). We focus on settings where the researcher is interested in a parameter other than the ATE in levels. In some settings, the researcher may be interested in the ATE in levels, but simple regression estimators may be noisy owing to a long right-tail of the outcome (Athey et al., 2021). The researcher might then try to estimate the ATE in levels by first estimating the ATE for a log-like transformation, and then multiplying by the baseline mean. However, since the ATE for a log-like transformation depends on the units of the outcome—and is thus not a true “percentage” effect—the validity of this approach for recovering the ATE in levels will depend on the initial units of Y .²¹ We refer the reader to Athey et al. (2021) and Müller (2023) for alternative approaches to estimation and inference targeted to settings where the ATE in levels is of interest but the outcome has heavy tails.

Remark 9 (Transformation-specific identification). Another reason that researchers may consider taking a transformation of the outcome is that a parametric assumption used for identification may be more plausible for some functional forms than others. For example, when the outcome is strictly positive, parallel trends in logs may be more plausible than parallel trends in levels if time-varying factors are thought to have a multiplicative impact on the outcome. We note that justifying parallel trends for a log-like transformation is especially tricky, however, since if parallel trends holds for the arcsinh of an outcome measured in dollars, say, it will not generally hold for the arcsinh of the outcome measured in cents (Roth and Sant’Anna, 2023). Thus, the parallel trends assumption is specific to both the transformation $m(\cdot)$ and the units of the outcome. Moreover, even if the researcher is confident in parallel trends for a particular log-like transformation and unit of the outcome, our results imply that they should not interpret the resulting ATT as an average

²¹Even in the case where Y is strictly positive and one first estimates the ATE in logs, this approach will only recover the ATE in levels under certain homogeneity assumptions, e.g. constant proportional effects. See Wooldridge (1992) for related discussion.